

Code Layout, Readability and Reusability

Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the codebase.	Yes	PEP-8 formatting followed.
2	Proper File Structure	Files and folders are logically organized.	Yes	templates/, static/, dataset/, model/ structure maintained.
3	Meaningful Variable Names	Variables clearly describe their purpose.	Yes	Descriptive names improve readability.
4	Function / Method Names	Functions are named according to their functionality.	Yes	Prediction and preprocessing functions are clearly defined.
5	Code Comments	Comments explain important logic and functionality.	Partial	Core functions and modules are documented.
6	Modular Design	Code is divided into reusable functions and modules.	Yes	Training, prediction, and preprocessing are separated.
7	No Redundant Code	Duplicate and unnecessary code is avoided.	Yes	Reusable functions reduce code duplication.
8	Error Handling	Invalid inputs and exceptions are handled gracefully.	Yes	Input validation and exception handling implemented.

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Crop Prediction Module	Python, Scikit-learn	Flask application and testing	High
2	Data Preprocessing Module	Pandas, NumPy	Training and prediction	High
3	Frontend Templates	HTML, CSS	Input and result pages	Medium
4	Model Loader	Pickle (.pkl)	Application startup	High
5	Visualization Module	Matplotlib	Data analysis and reports	Medium

Aspect	Rating(1-5)	Comments
Code Layout & Structure	5	Well organized
Readability	5	Easy to read.
Documentation/Comments	4	Important parts are commented.
Resusability	5	Reusable modules used.
Overall Score	4.8/5	Good quality code.

